

## cray1\_diagnostic\_sop\_v1.json

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```
{
  "id": "SOP-CRAY1-001",
  "title": "Cray-1: Processor Check and Error Correction Diagnostic",
  "system": "Cray-1",
  "component": "Central Processor / Memory SECDDED",
  "author": "SymbioStreams Semantic Bridge",
  "source": "Cray-1 Hardware Reference Manual (2240004C)",
  "steps": [
    {
      "step": 1,
      "action": "Indicator Analysis",
      "details": "When a 'Processor Check' occurs, examine the maintenance panel. Check if the floating point range error or vector error bit is set.",
      "indicator": "Check the status of the Exchange Package to see the program state at the time of error."
    },
    {
      "step": 2,
      "action": "Memory SECDDED Audit",
      "details": "Examine the syndrome bits for Single Error Correction Double Error Detection (SECDDED).",
      "check": "A single bit error is corrected automatically. A double bit error causes a system halt."
    },
    {
      "step": 3,
      "action": "Cooling System Verification",
      "details": "Verify liquid Freon coolant flow and temperature.",
      "logic": "If coolant temperature > 20C, the system may exhibit intermittent parity errors."
    }
  ]
}
```